

Objective

To implement and demonstrate `IllegalArgumentException` in Java using try-catch by providing an invalid thread priority value.

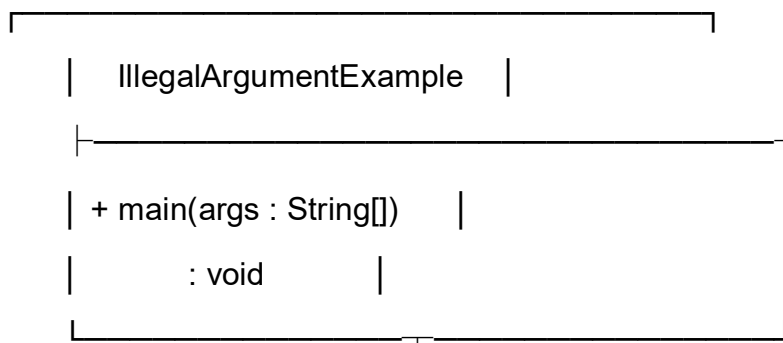
Steps for Execution

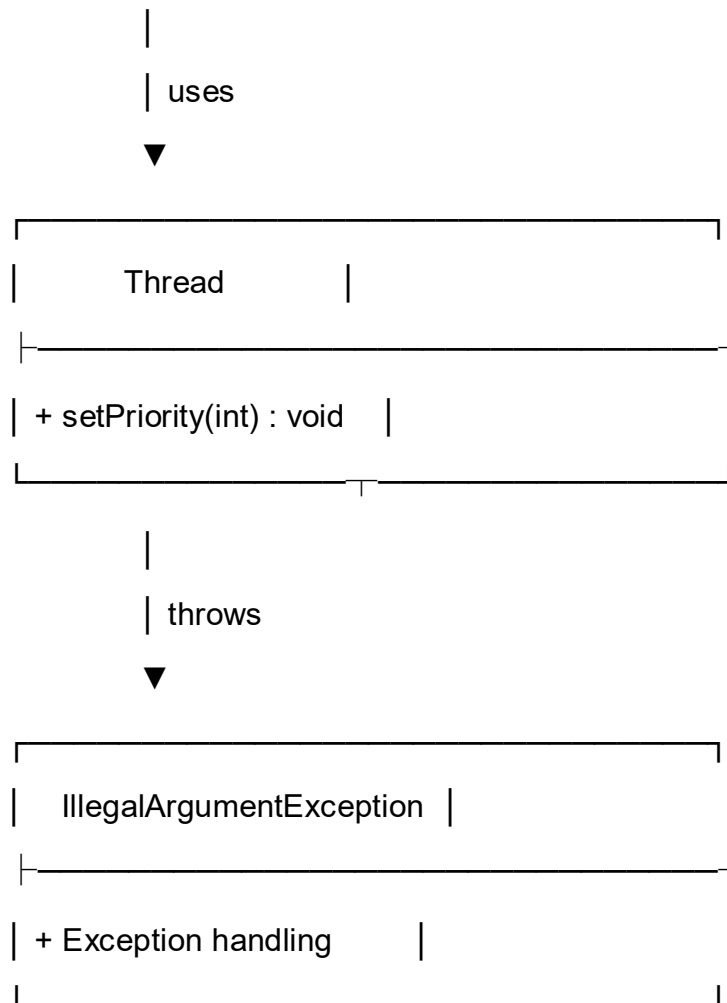
1. Create a Java project and package named `javacore`.
2. Create a file named `IllegalArgumentExceptionExample.java` and enter the given code.
3. Create a `Thread` object.
4. Try to set the thread priority to 15.
5. Handle the exception using catch.
6. Compile and run the program.
7. Observe the output.

Algorithm

1. Start the program.
2. Create a `Thread` object.
3. Try to set its priority to 15.
4. Catch `IllegalArgumentException`.
5. Display the error message.
6. Display "Program ended.".
7. Stop the program.

UML Class Diagram





Relationship

`IllegalArgumentException` creates and uses a `Thread` object.

`setPriority(15)` throws an `IllegalArgumentException`.

The exception is handled using the catch block.

Valid thread priorities range from 1 to 10.

CODE:

```
package javacore;

public class IllegalArgumentException {
    public static void main(String[] args) {

        Thread thread = new Thread();
```

```
try {  
    System.out.println("Setting thread priority...");  
  
    thread.setPriority(15);  
  
    System.out.println("Priority set successfully.");  
}  
catch (IllegalArgumentException e) {  
    System.out.println("Error: Priority must be between 1 and 10.");  
}  
  
System.out.println("Program ended.");  
}  
}
```

Output

Setting thread priority...

Error: Priority must be between 1 and 10.

Program ended.

Result

The program successfully demonstrates `IllegalArgumentException` handling in Java. Since the thread priority 15 is outside the valid range of 1 to 10, an `IllegalArgumentException` occurs and is handled using the catch block.